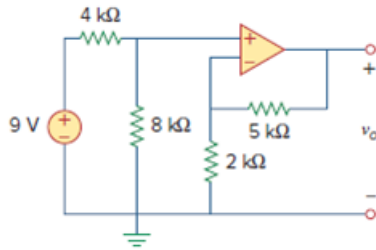


Problem

Calculate v_o in the circuit of Fig. 5.20.

Figure 5.20:

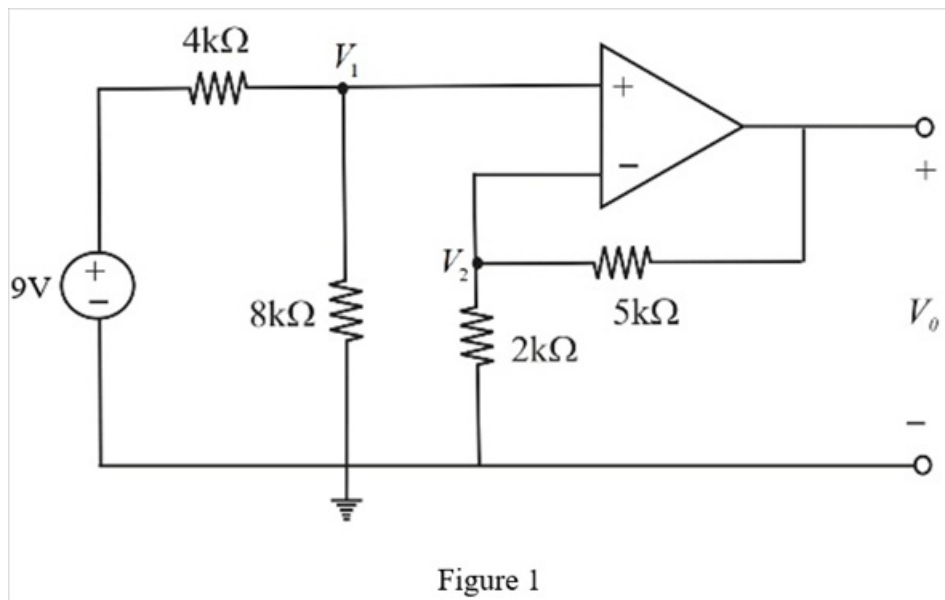


Step-by-step solution

Step 1 of 3

Refer to Figure 5.20 in the textbook.

Draw the circuit diagram with node voltages.



Step 2 of 3

Apply Kirchhoff's voltage law at node, V_1 .

$$\frac{V_1 - 9}{4k} + \frac{V_1}{8k} = 0$$

$$V_1 \left[\frac{1}{4k} + \frac{1}{8k} \right] = \frac{9}{4k}$$

$$V_1 [0.25 + 0.125] = \frac{9}{4}$$

$$V_1 [0.375] = \frac{9}{4}$$

$$\begin{aligned} V_1 &= \frac{9}{1.5} \\ &= 6 \text{ V} \end{aligned}$$

Apply Kirchhoff's voltage law at node, V_2 .

$$\frac{V_2}{2k} + \frac{V_2 - V_o}{5k} = 0$$

$$V_2 \left[\frac{1}{2k} + \frac{1}{5k} \right] = \frac{V_o}{5k}$$

$$0.7V_2 = 0.2V_o \dots\dots (1)$$

The Op-amp is ideal.

Therefore,

$$V_1 = V_2$$

$$\begin{aligned} V_2 &= V_1 \\ &= 6 \text{ V} \end{aligned}$$